

Cartostation - Towards the development of a convivial GIS

September 2025
FOSS4G Belgium

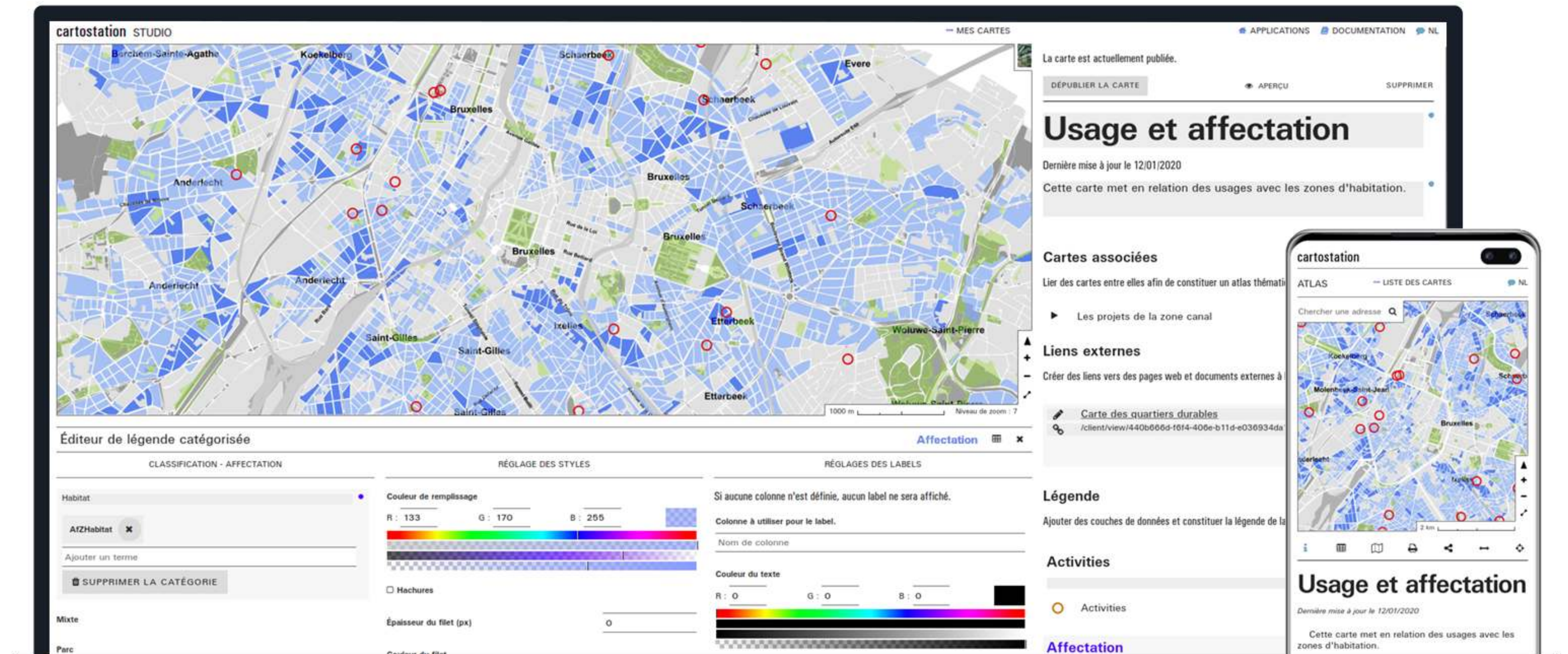
Cartostation

A modular and open-source webGIS focused on map publishing

- Crafted locally
- Originally dedicated to public organisations

Documentation & source code :

↳ [CARTOSTATION.COM](https://cartostation.com)



Cartostation - Technical stack

Relying on common technologies to facilitate contributions and collaboration.

- Python/Django
- React
- OpenLayers
- Typescript
- LESS-CSS

Cartostation - Main contributors and service providers

- Designed and primarily developed by Atelier Cartographique SC
- In regular collaboration with Champs Libres SCRLFS
- With more recent support from Tactic asbl
- And occasional contributions from other organisations

Atelier
carto
graphique

Champs  Libres

tac
tic

Atelier Cartographique SC

We are a self-managed workers' cooperative based in Brussels.

- Specialised in cartographic issues
- Focused on local actors
- We develop GIS softwares and provide cartographic support for public authorities, inter-community organisations, non-profit organisations, NGOs, and academic researchers
- We are active since 2013, and organised as a cooperative since 2021

↳ [ATELIER-CARTOGRAPHIQUE.BE](https://atelier-cartographique.be)

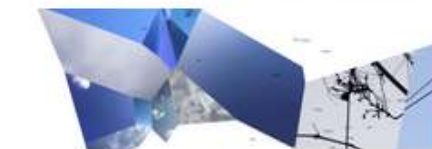
Liquid maps / Liquid stories
A trip to Namur with Buratinas



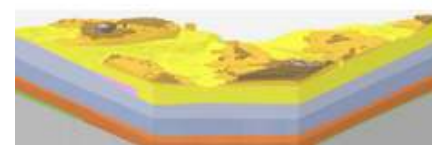
Silly
Maps and signs for soft mobility



Ciels
A map of skies on Waend



BrugeoTool
Mapping portal for geo-thermal potential in Brussels



Terre en vue
Online GIS & publishing platform



Eavatea
A map that is not a map



Infiltrasoil
Customised spatial reporting for rainwater infiltration in Brussels



Urban Plans and Projects monitoring platform
Instance of Cartostation for the public administration perspective.brussels



Wænd
Platform for collective cartographic experimentation



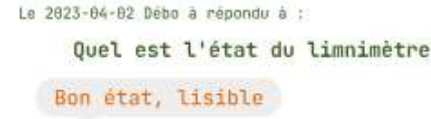
Tram81
Cartographic web application



SitEx
Survey application



SmartWater
An application for collecting data on water quality



Cartostation
A modular, open-source web GIS geared towards publishing maps



Carte de Brugelette
Printed map: OpenStreetMap data and participatory process



Cartographic design
Collaboration between EGBE and the Museum of Natural Sciences



The ExPEERience Map
Maps on human rights violations



osm.hiking.be
Hiking maps on demand



Qarto
OpenStreetMap data quality control tool for SNCF



Geodata
Spatial data infrastructure for the administration Brussels Environment



Doors and thresholds in the city of Mons
Cartographic web application



Simone Guillissen-Hoa
An illustrated leaflet for an exhibition



Hyphenated database
Digital memorial project



Rescue of the IGN sensitive archive
FRArt art research project



Atlas Wænd
Wænd as a publication



Maplog
Cartographic annotation



Topological Atlas
Navigation through research material on the construction of borders



Radio-robot
An experiment in data sound processing



Cartographic support
Cartographic support for the administration Perspective.brussels



Cartesolaire.brussels - Zonnekaart.brussels
Solar Cadastre of the Brussels-Capital Region



Buratinas
Cartographic web application



Atelier des Collectes
A methodology for collecting knowledge together



Cartofixer
Tools and public support



BruWater
Time series distribution platform



replyto.be
Public notebook



Woodwideweb
Web application and cartographic support for remarkable trees



Cartostation - Brief history (2016 → 2018)

Cartostation - Brief history (2016 → 2018)



CAHIER SPECIAL DES CHARGES

DU MARCHE PUBLIC DE

SERVICES

AYANT POUR OBJET

**"INFRASTRUCTURE DE DONNÉES
SPATIALES"-2016D0603**

PROCÉDURE NÉGOCIÉE SANS PUBLICITÉ

Selon l'article 26, § 1, 1° a

Pouvoir adjudicateur

**Institut Bruxellois pour la Gestion de
l'Environnement**

Auteur de projet

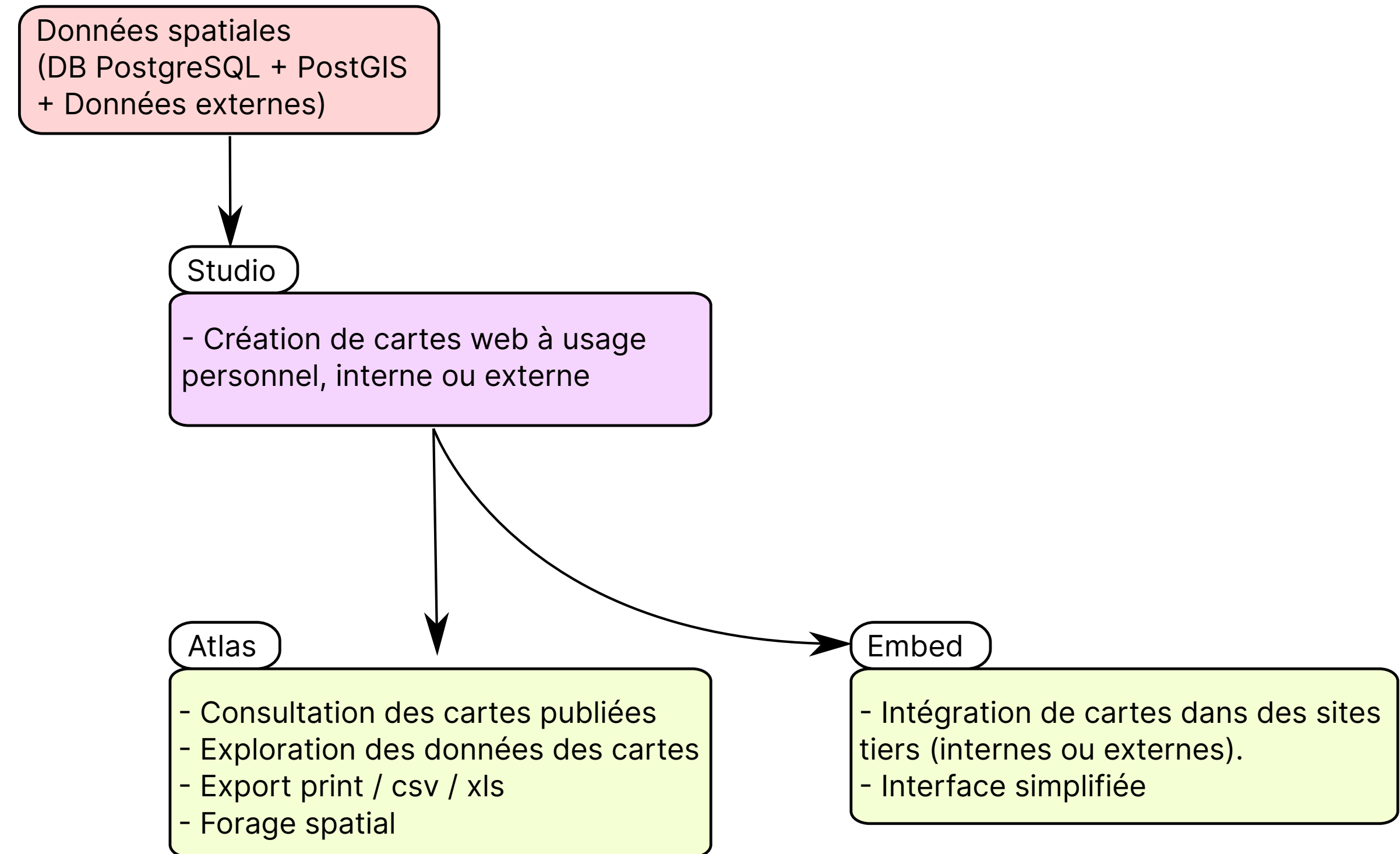
**Division Facilities et Patrimoine Immobilier, Gael Kruwialis
Avenue du port 86C/3000 à 1000 Bruxelles**

Cartostation - Modularity & Map Publishing

From spatial data to online map publishing

The core of Cartostation:

- A spatial database (PostgreSQL + PostGIS), potentially supplemented by external data
- **Studio**: A module for creating and publishing maps
- **Atlas**: A feature-rich map viewer
- **Embed**: A simplified viewer for integrating maps into third-party websites



Cartostation - 'Studio' application

A tool to facilitate the production of high-quality web maps, both visually and editorially.

- Extensive editorial capabilities
 - Map title, description and illustration
 - Attached files, related links
 - Related maps / Micro-Atlas
- Continuous and Discret categories
- Data formating
- Publishing control
- Version management
- Background maps subset per map
- ... among other features



And within an interface that guides you through the important steps of creating a map.

cartofixer > Studio

Mes cartes

carto_admin NL

Infos Légende Fiches Relations Publication

Légende

Ajouter des couches de données et constituer la légende de la carte.

+ AJOUTER UNE COUCHE DE DONNÉES

Café des savoirs 18/06/2022

Résultats du café des savoirs

- Hydrologie
- Histoire(s) locale(s)
- Points de tensions
- Biodiversité
- Projets réalisés, en cours ou planifiés
- Idées et visions
- Initiatives, dynamiques citoyennes liées à l'eau
- Initiatives, dynamiques citoyennes liées à l'environnement

Topographie (m-DNG)

Topographie (m-DNG)

SYMBOL

Label de légende (facultatif)

Résultats du café des savoirs

Hydrologie

Histoire(s) locale(s)

Termes filtrés

Histoire(s) locale(s)

Sélectionner une valeur

Points de tensions

Biodiversité

Projets réalisés, en cours ou planifiés

Idées et visions

Initiatives, dynamiques citoyennes liées à l'eau

Initiatives, dynamiques citoyennes liées à l'environnement

+ AJOUTER UNE CATÉGORIE

RÉGLAGE DES STYLES

RÉGLAGES DES LABELS

Si aucune colonne n'est définie, aucun label ne sera affiché.

Nom de colonne

Couleur du texte

Teinte

Opacité

Corps du texte (px)

Position du label

Offset X

Offset Y

Zoom min

cartostation

Cartostation - 'Atlas' application

A thematic atlas and comprehensive map viewer

- Browse published maps according to your user role permissions
- Filter maps by theme
- Search maps by keyword

geodata.environnement.brussels

ATLAS

Menu

Se connecter

NL

Bienvenue sur l'atlas

Découvrez ici les applications et les cartes interactives de Bruxelles Environnement

Chercher dans l'atlas

Filtrer par thématique

☐ Eau (14)

☐ Sol et Sous-sol (10)

☐ Espaces verts et biodiversité (24)

☐ Bruit (9)

☐ Air et Climat (2)

☐ Permis et inspection (3)

☐ Ville en transition (10)

☐ Loisirs (2)

☐ Bien-être animal (2)

☐ Mobilité (3)

☐ Bâtiments et énergie (2)

Pour accéder à nos données (en webservice), consultez notre [site](#)

déclaration d'accessibilité

contactez-nous

L'atlas des cartes

Territoire couvert par la Zone de basses émissions (LEZ)
Cette carte indique le territoire couvert par la Zone de Basses Emissions (LEZ) de la Région de Brux...

Inondation aléa et risque
Cette carte d'aléa d'inondation (de 2019) repère les zones où pourraient se produire des inondations...

Zones d'accessibilité (RRU)
Zones d'accessibilité en transport en commun selon le Règlement Régional d'Urbanisme (RRU). La Régio...

Potagers de Bruxelles Environnement
Sur les 14 sites qu'il gère, Bruxelles Environnement met ainsi à disposition des jardiniers amateurs...

Habitats Natura 2000
Protection de la biodiversité: Habitats Natura 2000. Habitats d'intérêt communautaires et régionales...

Biotopes de haute valeur biologique
Cette carte montre les biotopes de haute valeur biologique dans la Région bruxelloise. La carte est ...

Amphibiens et reptiles en Région bruxelloise
Répartition des amphibiens et reptiles en Région de Bruxelles-Capitale

Sites nature
Protection de la biodiversité. Sites qui bénéficient d'une protection nature: Sites Natura 2000, rés...

Espaces verts accessibles au public
Retrouvez les espaces verts accessibles au public en fonction de différentes topologies.

Etat du sol
La carte de l'état du sol reprend toutes les parcelles cadastrales pour lesquelles Bruxelles Environ...

Antennes émettrices
La carte reprend les antennes soumises à permis d'environnement ainsi que les principales données te...

Stratégie zones de confort acoustique
Afin de répondre aux obligations de la Directive 2002/49/CE relative à l'évaluation et à la gestion ...

Niveaux de bruit multi-exposition
La multi-exposition recouvre le bruit routier, ferroviaire et aérien.

Niveaux de bruit du trafic aérien

Niveaux de bruit du trafic ferroviaire

Niveaux de bruit du trafic routier

Potagers collectifs et familiaux (2018)

Végétation 2021
Répartition de la végétation haute et basse en sep...

Réseau écologique bruxellois
Le réseau écopro...

Carte d'évaluation biologique
L'ÉGÉNDE CI-DESSOUS ...

Arbres dans les parcs régionaux
Répartition des arbres gé...

cartostation

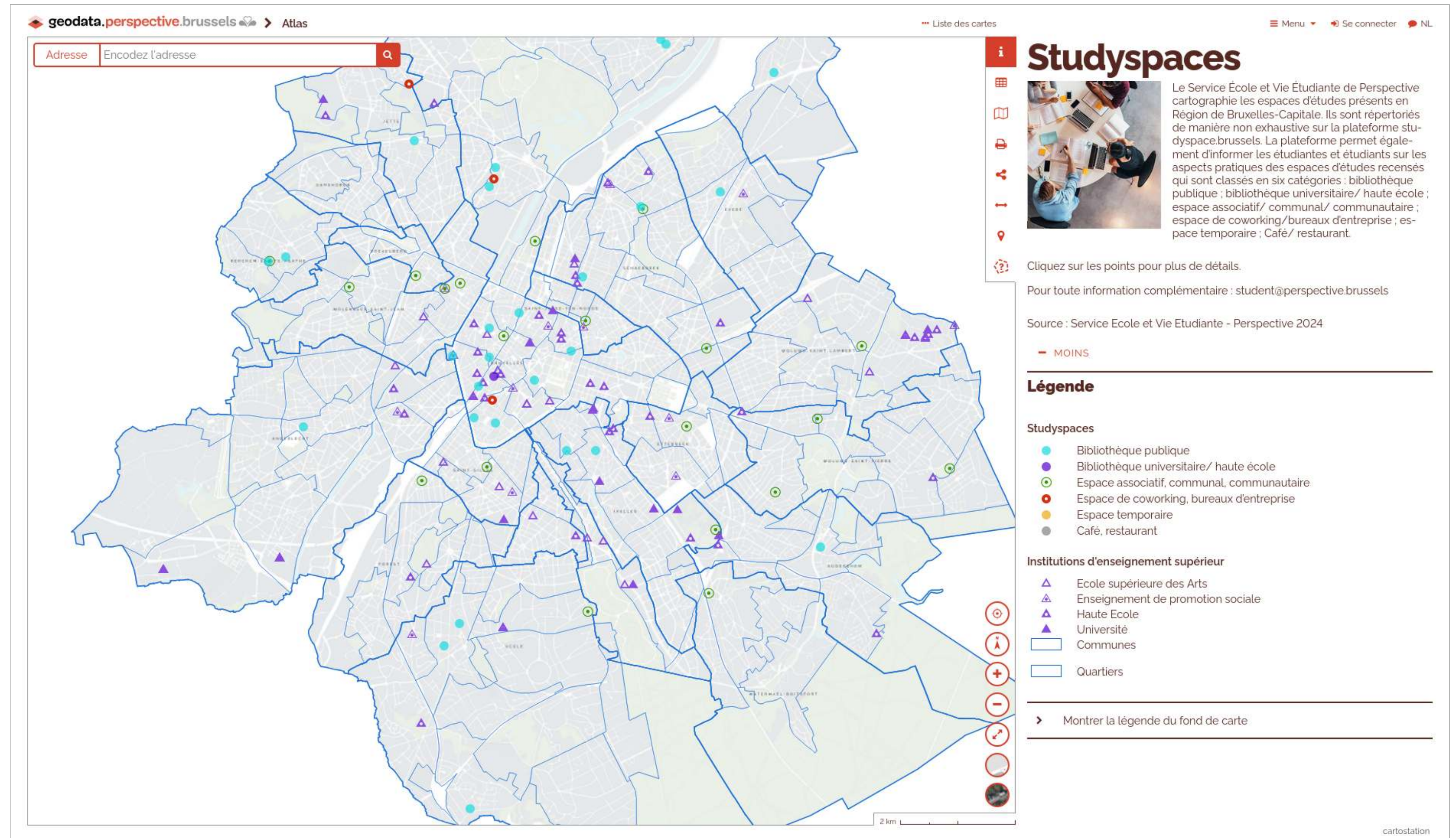
Atelier
carto
graphique

Cartostation - 'Atlas' application

A thematic atlas and comprehensive map viewer

- Title, description, Legend
- Data exploration (attribute table, filters, exports)
- Background-map selector
- Map PDF exports
- Sharing (url and embed code w/ current bounding box)
- Measurement tools
- Spatial filters, per layer and related maps
- Geocoder (Urbis, nominatim, NOVA, Capakey, ...)
- Map element query
- Geolocation
- Spatial bookmarks

(+ data-streaming capabilities, so you can mobilise > 100k features in a map or a table without melting your browser - Cadaster and Buildings, we are looking at you !)

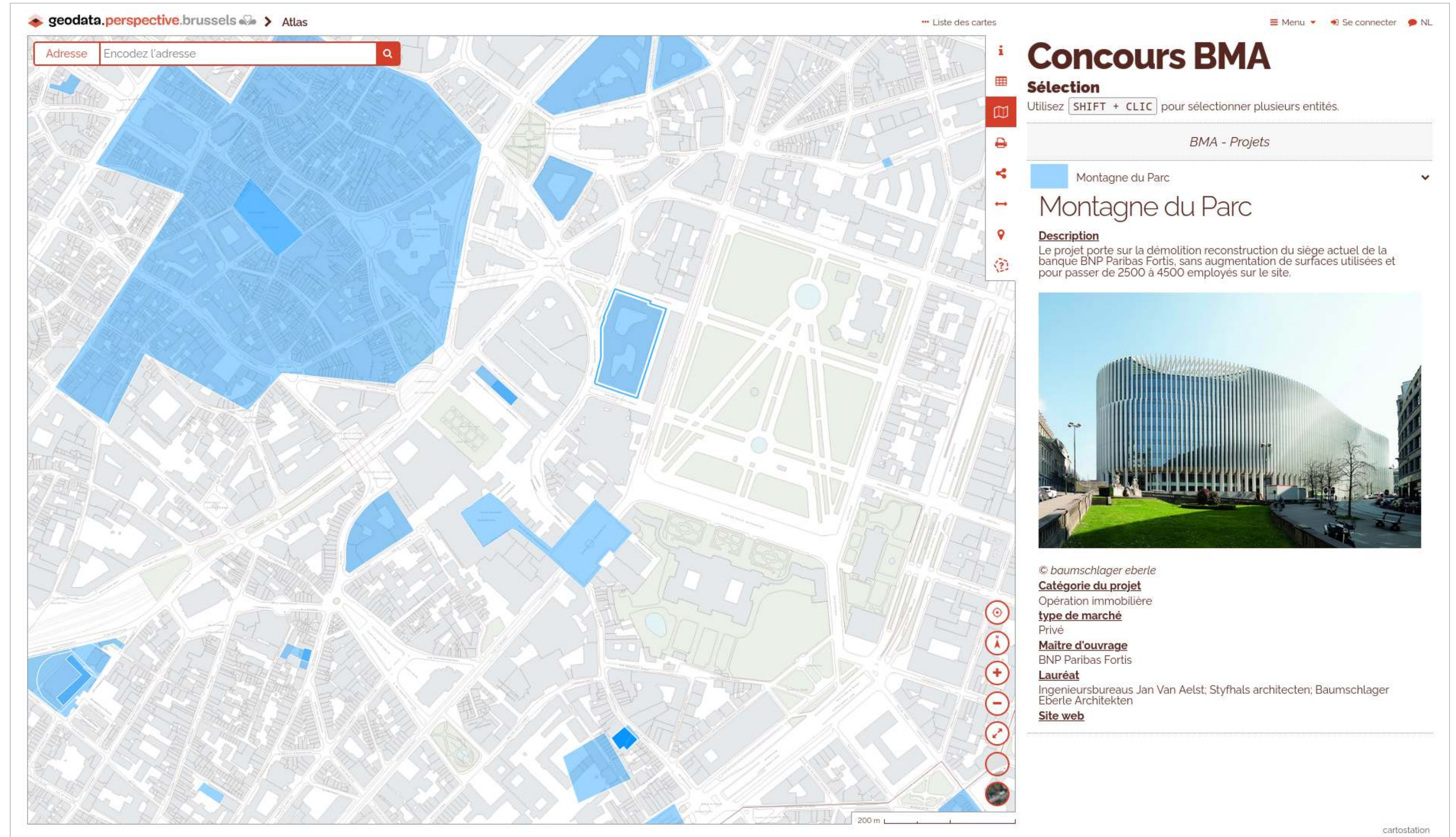


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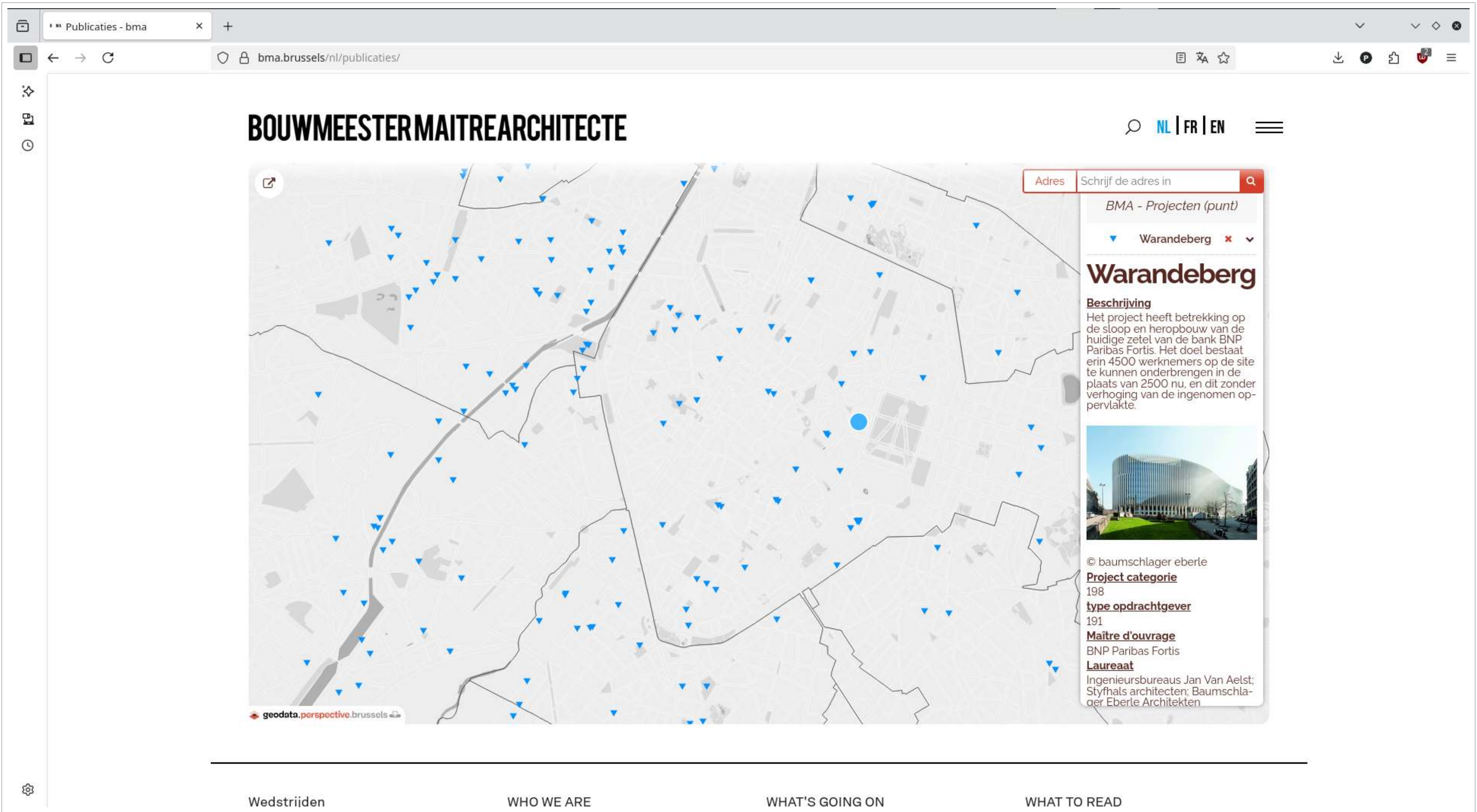
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Cartostation - 'Embed' application

A simple viewer to embed maps on a third-party website

- Simplified map viewer for integrating interactive maps into third-party websites.
- Several viewer templates available



↳ EMBEDDED MAP - BMA.BRUSSELS

Cartostation - since 2018

Strengthening the key components, developing new specific modules, and addressing more in-depth issues that arise.

- Cartostation as a framework to develop dedicated applications
- Multiple experiences in various contexts
- > 10 Cartostation instances that we know of

SitEx
Survey application



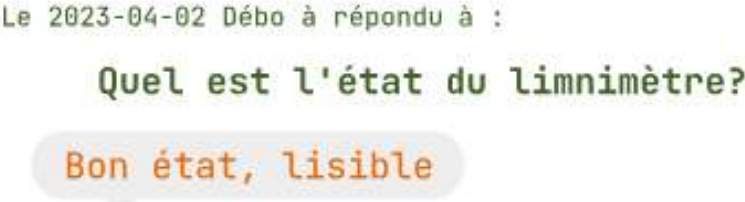
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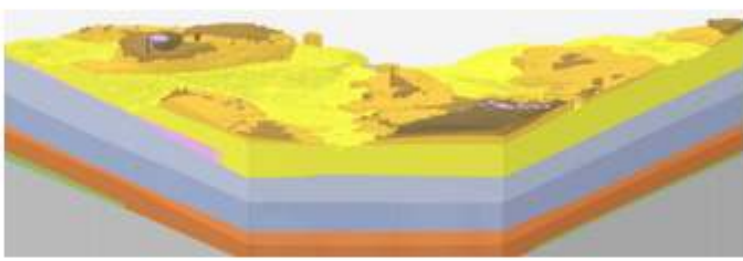
Urban Plans and Projects monitoring platform
Instance of Cartostation for the public administration perspective.brussels



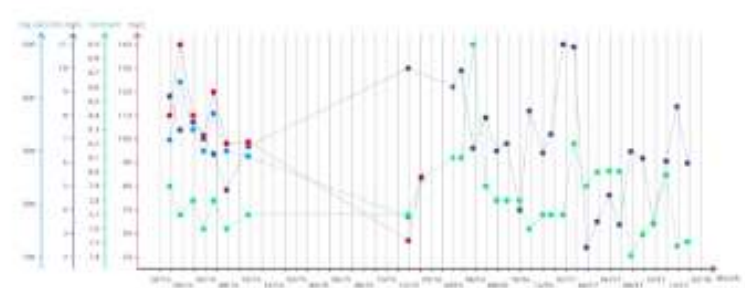
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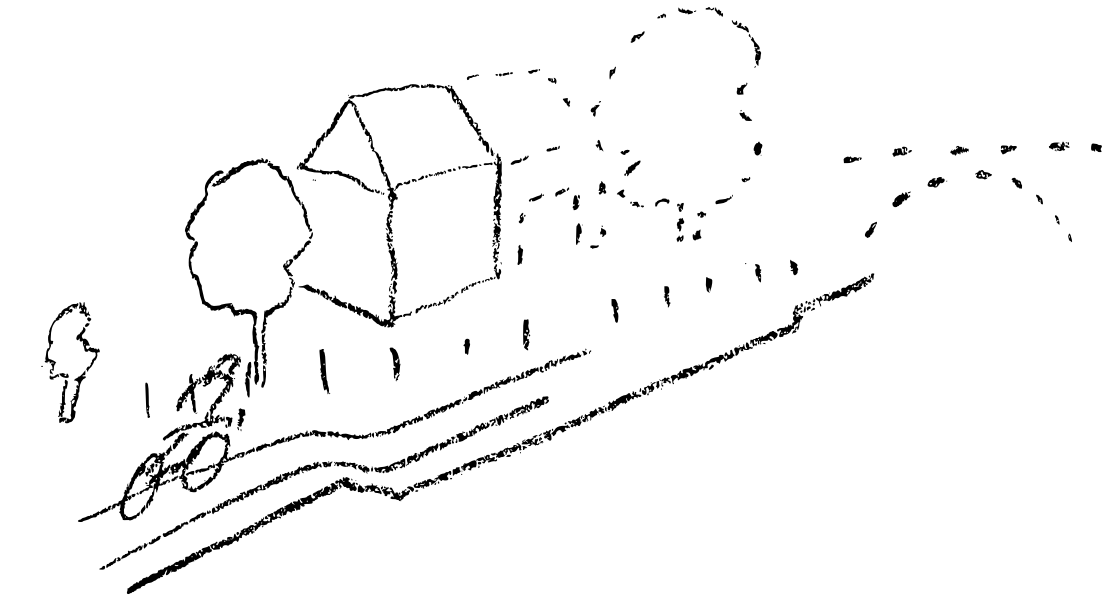
Geodata
Spatial data infrastructure for the administration Brussels Environment



Cartostation - Other topics that matter : Data production

The multiple approaches to spatial data production

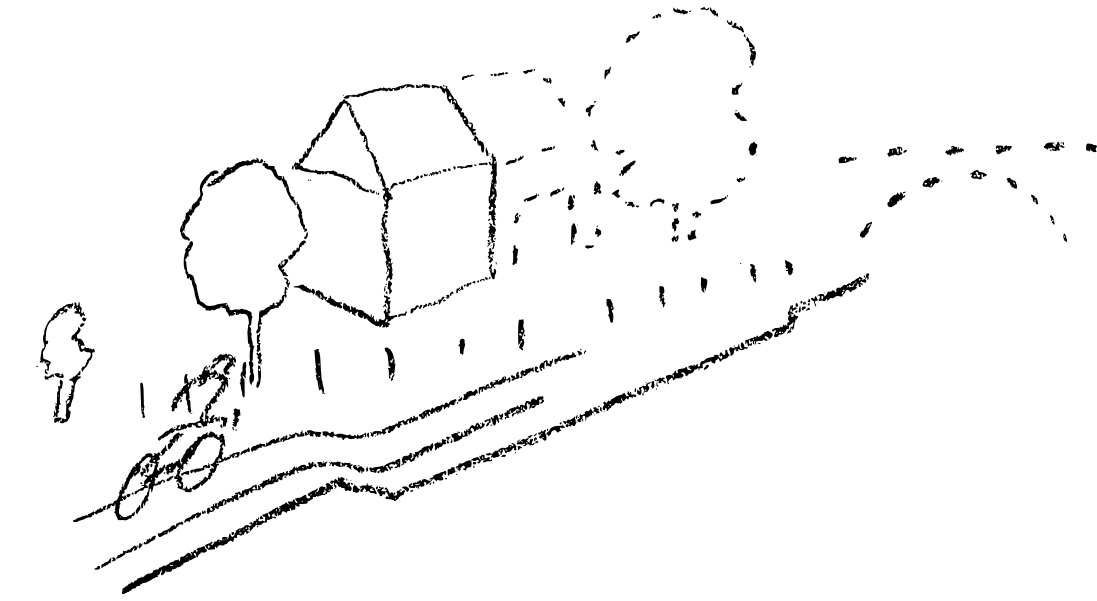
- Long-term monitoring of urban elements



Cartostation - Other topics that matter : Data production

The multiple approaches to spatial data production

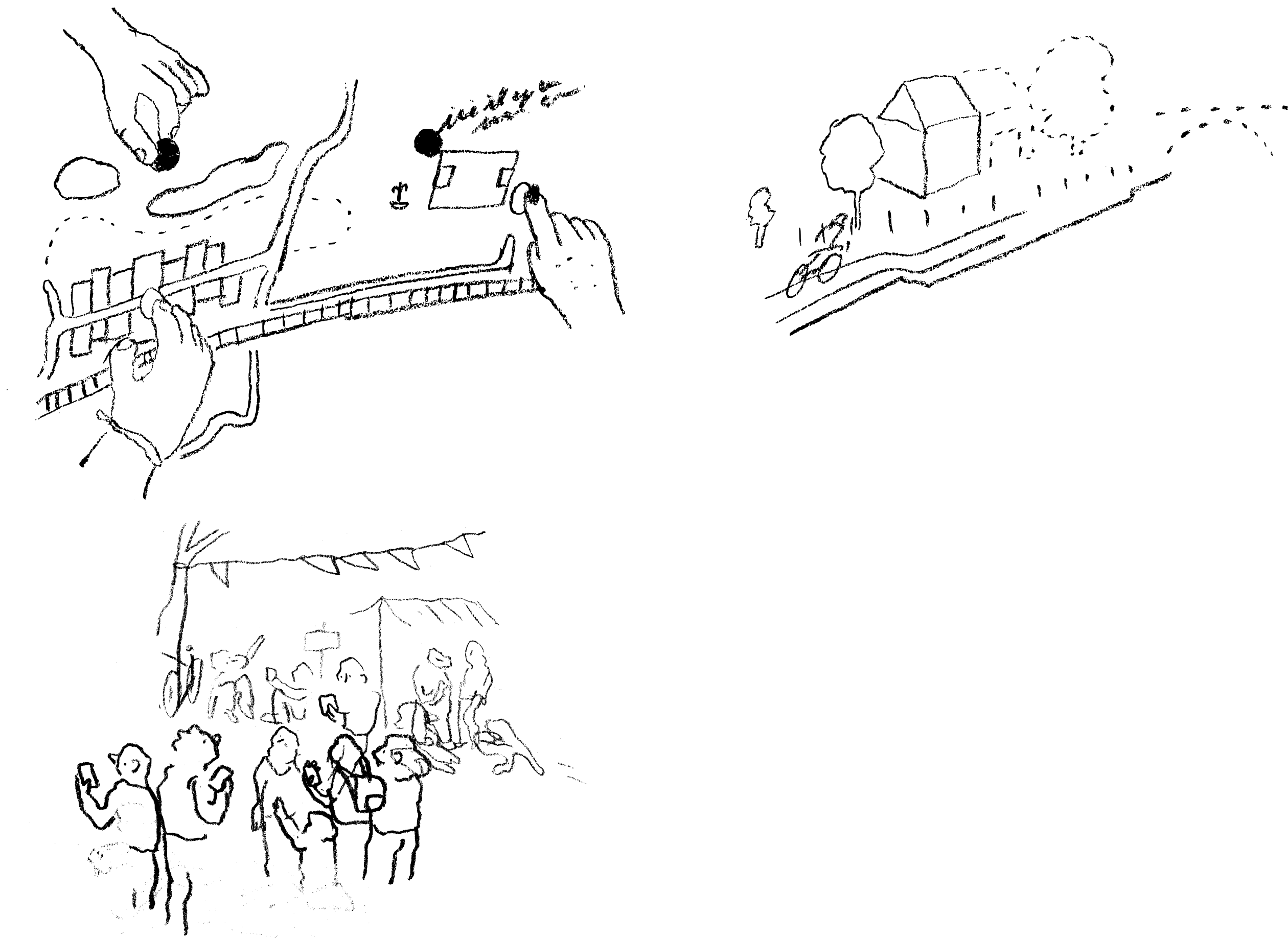
- Long-term monitoring of urban elements
- Data collection by citizens



Cartostation - Other topics that matter : Data production

The multiple approaches to spatial data production

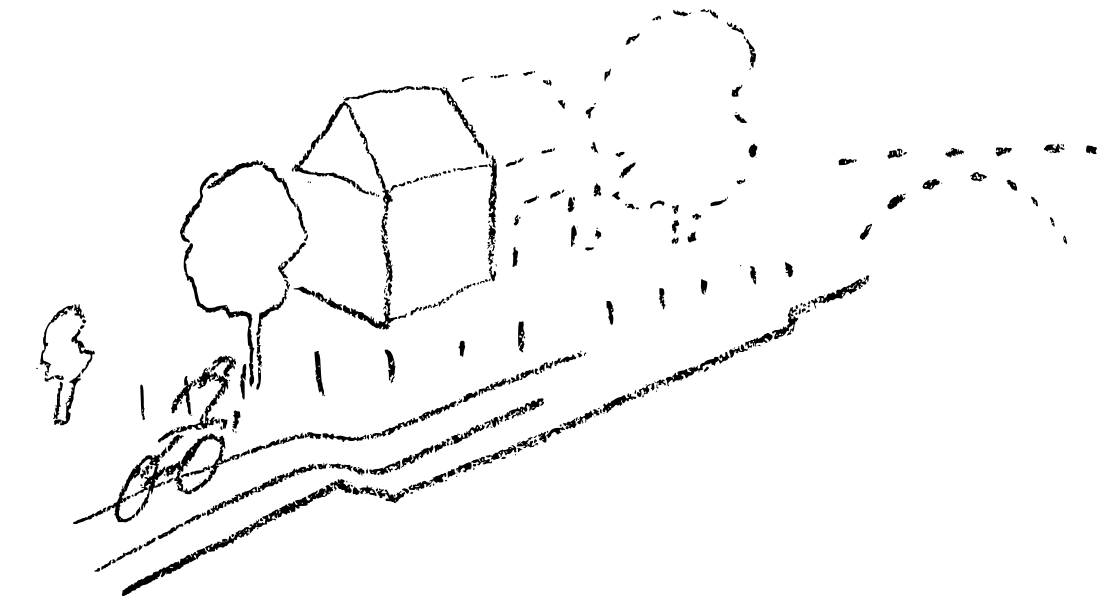
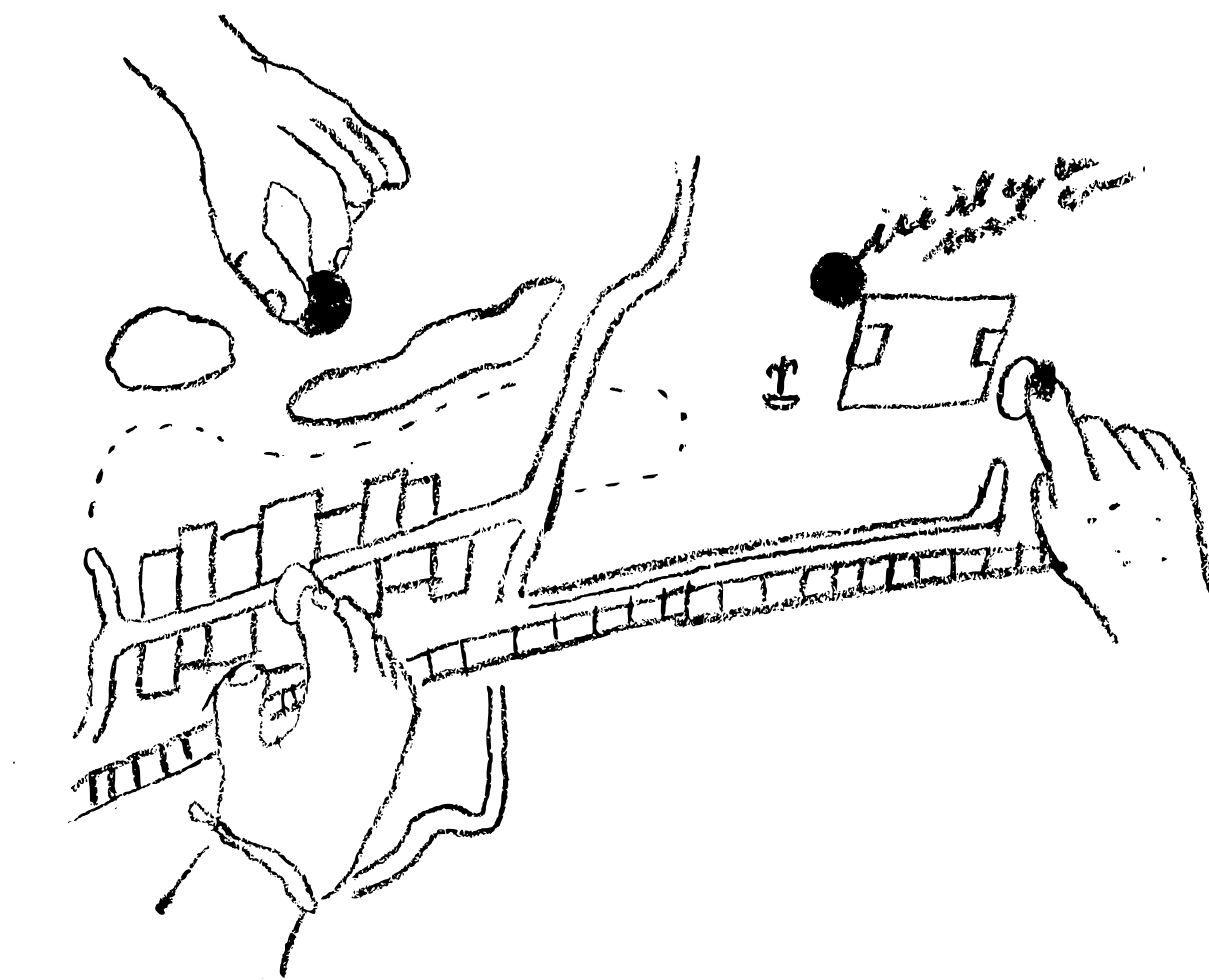
- Long-term monitoring of urban elements
- Data collection by citizens
- Cartographic annotation



Cartostation - Other topics that matter : Data production

The multiple approaches to spatial data production

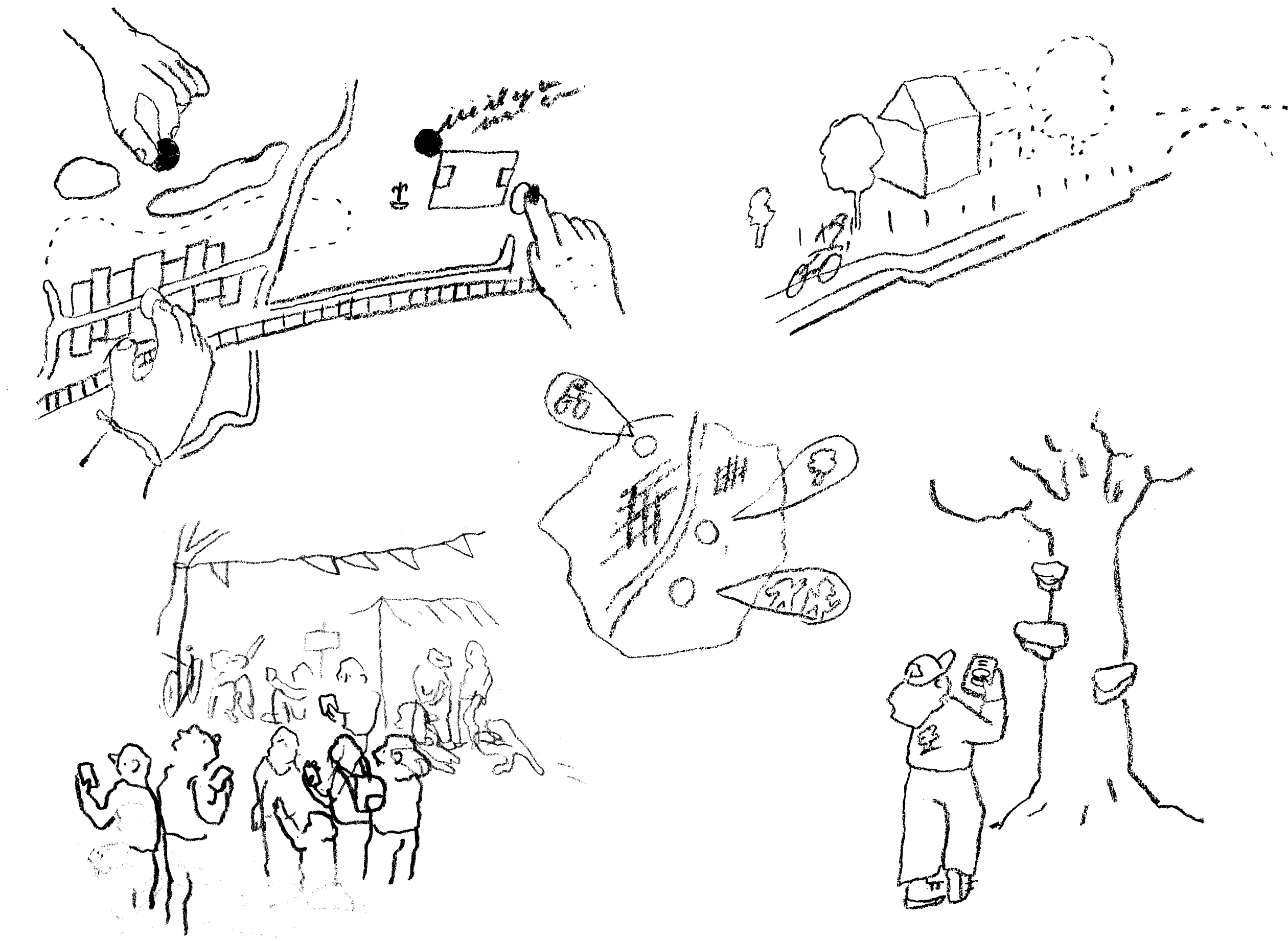
- Long-term monitoring of urban elements
- Data collection by citizens
- Cartographic annotation
- Field surveying campaigns



Cartostation - Other topics that matter : Data production

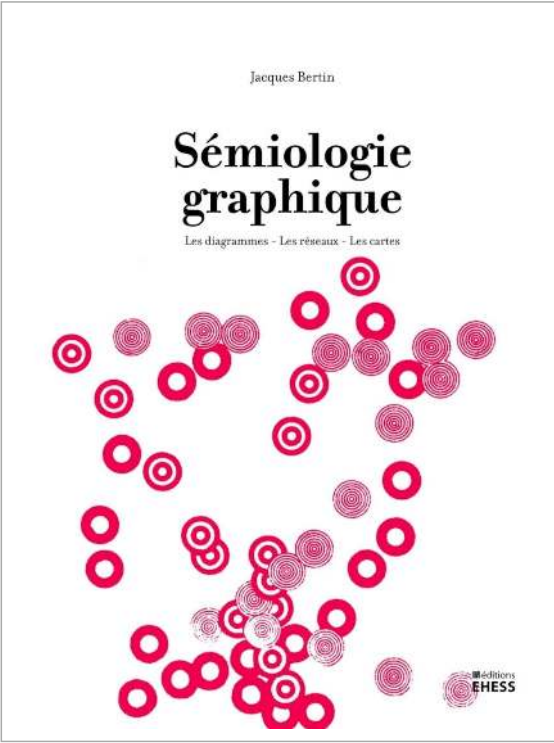
The multiple approaches to spatial data production

- Long-term monitoring of urban elements
- Data collection by citizens
- Cartographic annotation
- Field surveying campaigns
- etc.



Cartostation - Other topics that matter : Cartography

On aesthetic, typographical, and cartographic issues

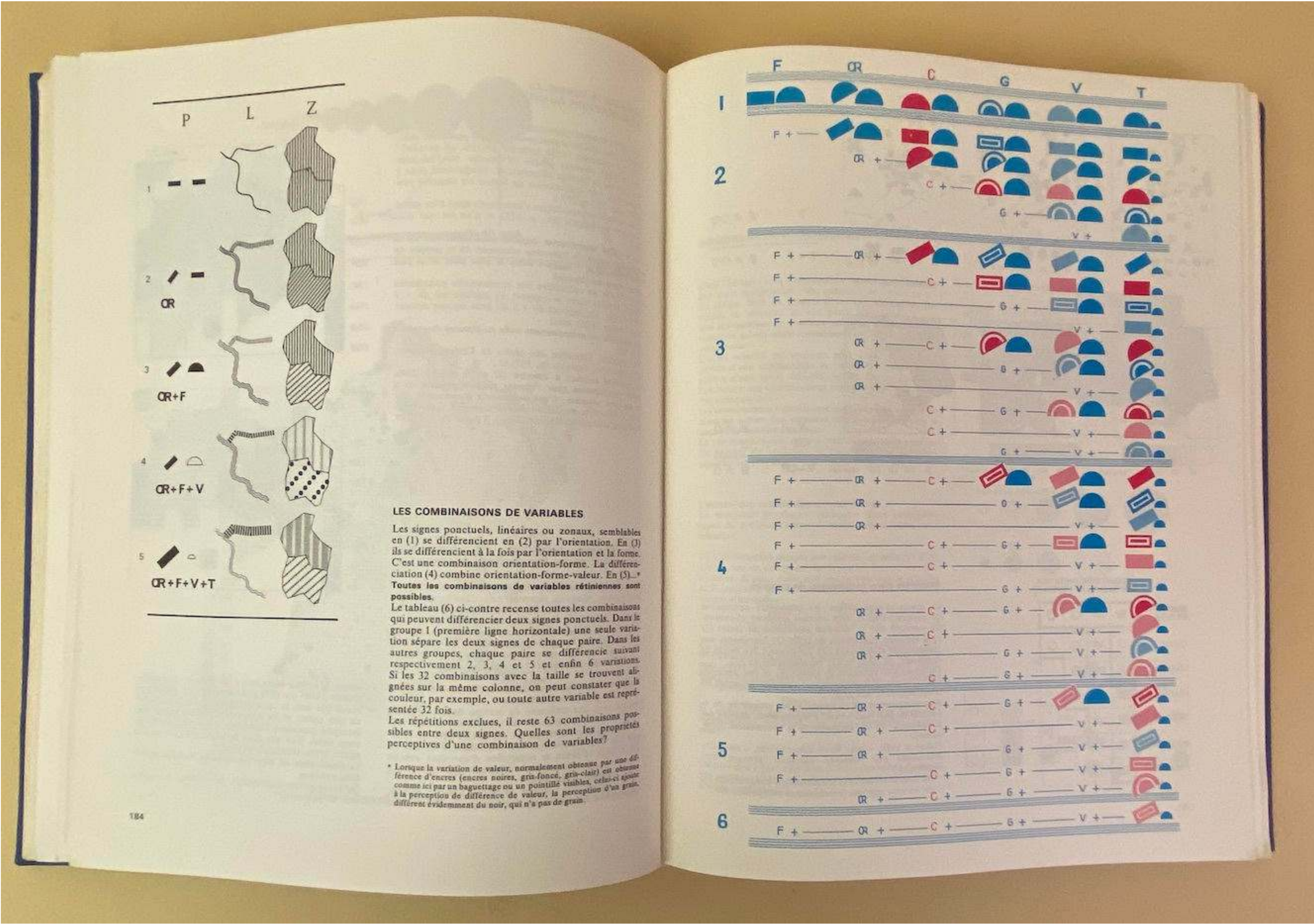
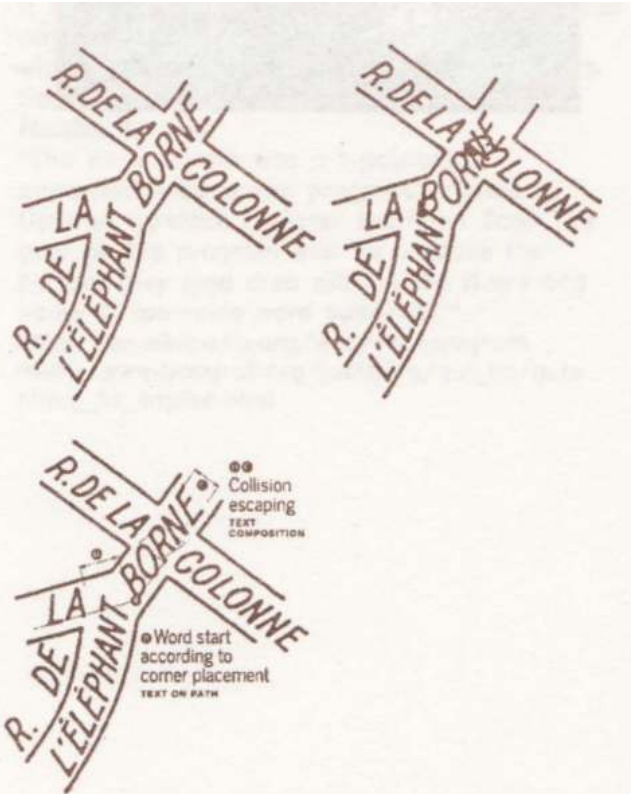


you understand how this can result for maps.

It? an engine that can combines g and visual intelligence? How could human intelligence; a form of ation that can be used in different

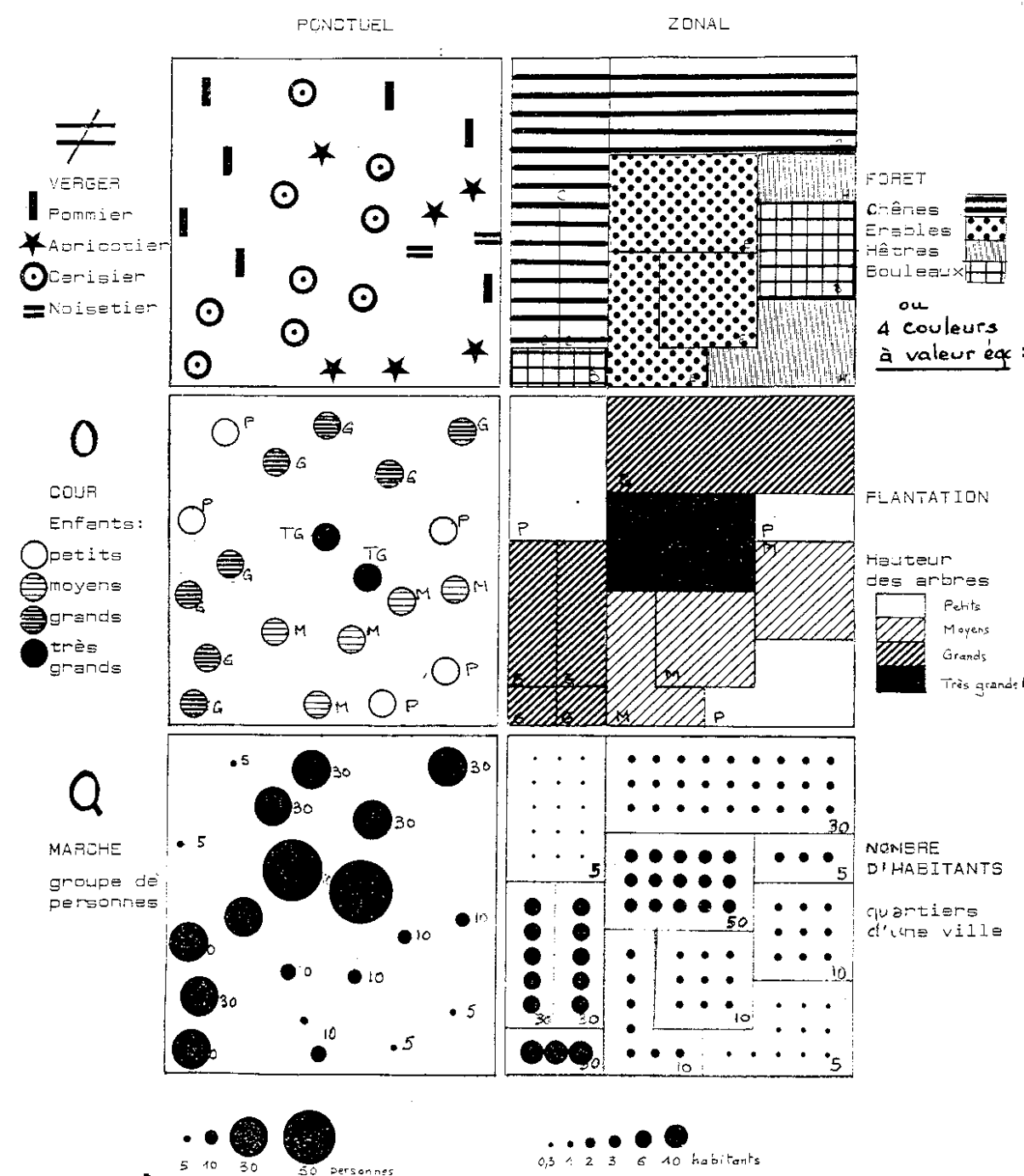
g-flickr.net/en/2012/06/28/ng-all-new-up-to-date-maps ptimization Collision

Googlemaps and OpenStreetMaps itations: How are decisions made? uments, parameters are taken into



Cartostation - Other topics that matter : Cartography

On aesthetic, typographical, and cartographic issues



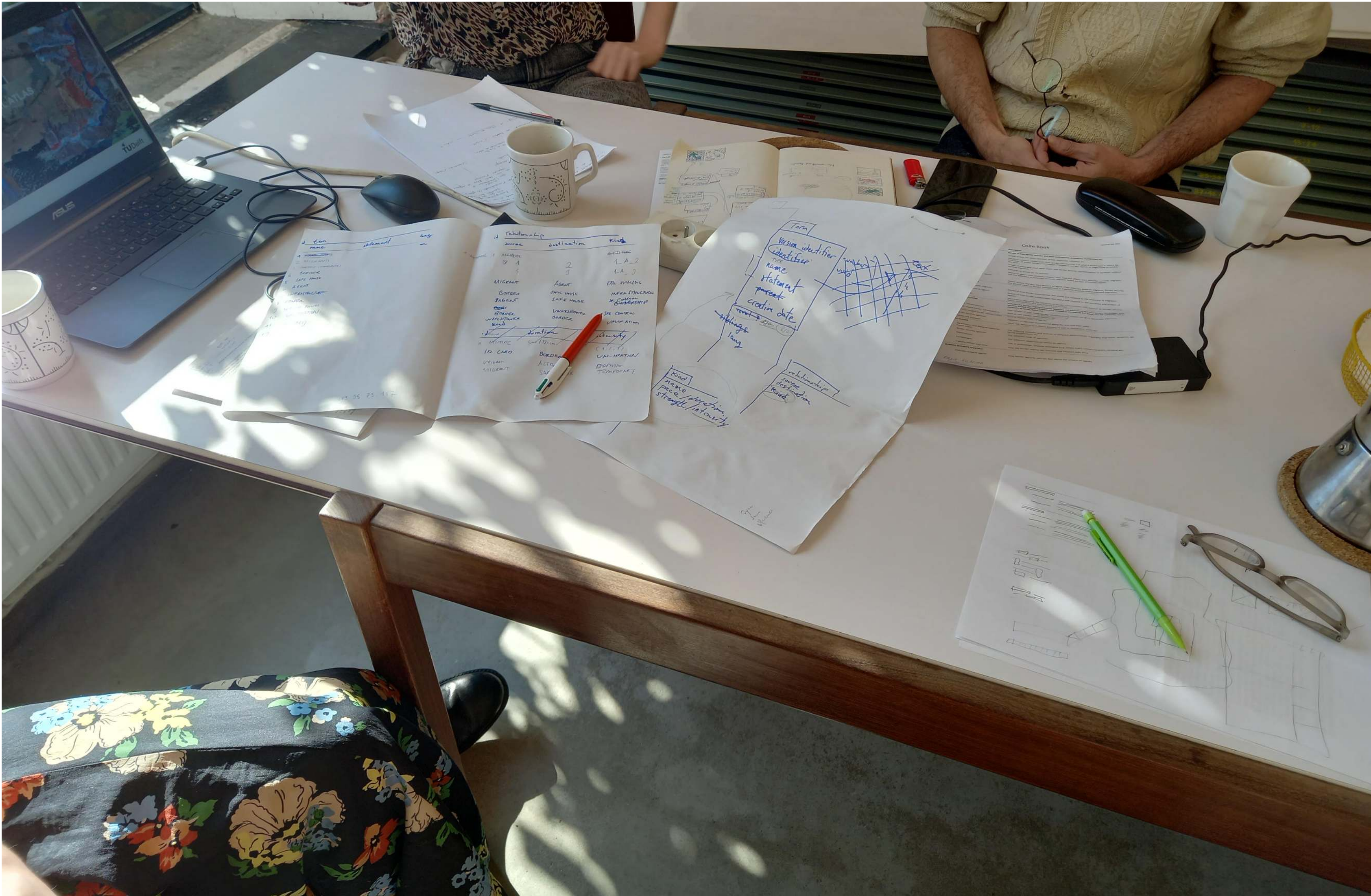
Cartostation - Other topics that matter : Common languages

On the creation and maintenance of common languages, in order to be able to name and structure information on a common basis.



Cartostation - Other topics that matter : Community

On the relevance – and complexity – of pooling needs and resources



Cartostation - Other topics that matter : Conviviality

Towards a combination of software development and cross-cutting analysis and support, which engages in dialogue with other issues at stake.

« J'entends par *convivialité* l'inverse de la productivité industrielle. Chacun de nous se définit par relation à autrui et au milieu et par la structure profonde des outils qu'il utilise. Ces outils peuvent se ranger en une série continue avec, aux deux extrêmes, l'outil dominant et l'outil convivial. »

Ivan Illich - *La convivialité* - 1973

«I choose the term “conviviality” to designate the opposite of industrial productivity. I intend it to mean autonomous and creative intercourse among persons, and the intercourse of persons with their environment; and this in contrast with the conditioned response of persons to the demands made upon them by others, and by a man-made environment. I consider conviviality to be individual freedom realized in personal interdependence and, as such, an intrinsic ethical value. I believe that, in any society, as conviviality is reduced below a certain level, no amount of industrial productivity can effectively satisfy the needs it creates among society's members.»

Ivan Illich - *Tools for Conviviality* - 1973

Ivan Dominic Illich (4 September 1926 – 2 December 2002) was an Austrian Catholic priest, theologian, philosopher, and social critic.

Join the conversation

Let's embed this approach in the development of Cartostation as an open-source project: not only as software, but also as a way of implementing, activating and developing it within local communities that bring together producers and users.

